

Lab 2 Tasks – Section G

Task 1: Identify and Label Network Hardware with Protocol Analysis (2 Marks)

For this task, you will open Cisco Packet Tracer and add several devices to your workspace. Begin by adding the following devices:

- **One 1841 Router** to serve as the central routing device for interconnecting networks.
- **One 2960-24TT Switch** to create the central hub for connecting devices within the same network.
- **One Generic Hub** to demonstrate a simple LAN connection where all devices are connected to a single device that broadcasts data to all.
- **Two PC-PT devices**, which will represent end-user computers in the network, capable of sending and receiving data.
- **One Laptop-PT** for wireless connectivity to the network, representing a portable device.
- **One Wireless Router-PT** to act as the source of the wireless network, connecting wireless devices to the local network.

Once the devices are added, each device should be labeled with its name and a brief description of its purpose. For example, the router connects different networks, while the switch connects devices within the same network. Take a screenshot of the entire network and ensure each device is clearly labeled with its name and function.

Task 2: Create Network Topologies with Protocol Layer Simulation Analysis (4 Marks)

For this task, you will simulate different network topologies using Cisco Packet Tracer with **protocol analysis**. Each topology should be configured with four PC-PT devices and appropriate network structure.

1. Star Topology with Simulation Mode Analysis:

Create a star topology where all four PCs are connected to a central 2960-24TT Switch. Assign IP addresses:

- PC1: 192.168.1.1
- PC2: 192.168.1.2
- PC3: 192.168.1.3
- PC4: 192.168.1.4

Protocol Analysis Steps:

- **Enable Simulation Mode** and filter for **ICMP** and **ARP** protocols
- From PC1: ping 192.168.1.3 -n 1
- **Step through simulation** and examine:
 - **ARP Request/Reply** packets (if needed for MAC address resolution)

- **ICMP Echo Request** packet details (Type, Code, Sequence number)
- **ICMP Echo Reply** packet details
- **Frame encapsulation** at each OSI layer
- **Document:** How the switch handles unicast traffic vs broadcast traffic

2. Ring Topology:

Create a ring topology where four PCs are connected in a circular fashion (using 4 switches).

Assign IP addresses:

- PC1: 192.168.2.1
- PC2: 192.168.2.2
- PC3: 192.168.2.3
- PC4: 192.168.2.4

Use ping command to test connectivity and observe data flow patterns.

3. Bus Topology:

Create a bus topology with 4 devices (using switches to simulate). Assign IP addresses:

- PC1: 192.168.3.1
- PC2: 192.168.3.2
- PC3: 192.168.3.3
- PC4: 192.168.3.4

Test connectivity using ping command.

4. Mesh Topology with Traceroute Analysis:

Create a mesh topology where each PC is connected to every other device. Assign IP addresses:

- PC1: 192.168.4.1
- PC2: 192.168.4.2
- PC3: 192.168.4.3
- PC4: 192.168.4.4

Analysis:

- Use ping command to test basic connectivity
- Use tracert command from PC1 to PC4 to observe the direct path
- **Compare** with a simulated "longer path" scenario
- **Document:** How mesh topology provides redundant paths

For each topology, document one advantage or disadvantage based on your **protocol-level observations** from simulation mode.

Task 3: Wireless Network with Multi-Network Traceroute Analysis (4 Marks)

Network Setup:

1. Add Devices:

- One WRT300N Wireless Router (centrally located)
- Three PC-PT devices positioned within wireless range
- Configure all PCs with wireless modules (Physical tab: remove Ethernet, add WMP300N)

2. IP Configuration:

- PC1: 192.168.0.10/24, Gateway: 192.168.0.1
- PC2: 192.168.0.11/24, Gateway: 192.168.0.1
- PC3: 192.168.0.12/24, Gateway: 192.168.0.1
- WRT300N: 192.168.0.1/24

3. Wireless Router Configuration:

- SSID: "cnet_lab2"
- Security: WPA2-PSK
- Password: "lab2secure"
- DHCP: Disabled

Tasks:

A. Wireless Network Setup:

- Configure WRT300N with specified SSID and security settings
- Connect all three PCs to wireless network
- Assign static IP addresses as specified
- Take screenshot of network topology with wireless connections

B. Connectivity Testing:

- PC1 ping PC2 (192.168.0.11)
- PC1 ping PC3 (192.168.0.12)
- PC2 ping PC3 (192.168.0.12)
- All PCs ping router (192.168.0.1)

C. Traceroute Analysis:

- From PC1: tracert 192.168.0.11
- From PC1: tracert 192.168.0.12
- From PC2: tracert 192.168.0.12

Submission Guidelines:

- Submit a single PDF document containing all tasks and the CISCO packet files to the GCR, **ensure your file name is visible in all pictures and contains your roll number.**
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **J_i221234_Lab1.**
- Name the CISCO files as follows: SectionLetter_RollNumber_TaskNumber, for example: **J_i221234_T1.**
- In the demo, there will be deductions based on your answers to the questions asked.

NOTE:

- You must show clear screenshots of every step you take. Marks will be deducted for this mistake and in some cases you may be awarded 0 (this applies to all tasks).
- Screenshots where your name, roll number and section are NOT visible, will automatically be given 0.

Sample screenshot:

